PROJECT DATA SHEET

CASE Consultants
Civil And Structural Engineers
Moose Hall
Barewell Road
Torquay
TQ1 4PA
tel. (01803) 200175
fax. (01803) 323205
E-Mail: admin@caseconsultants.net

JOB/PROJECT : NOSS ON DART MARINA, DARTMOUTH

JOB/PROJECT NO : C8829 **SHEET NO :** COVER

GENERAL :

CLIENT : PREMIER MARINAS

ARCHITECT : N/A

AUTHORITY : TORBAY COUNCIL

JOB/PROJECT DESCRIPTION : KING POST RETAINING WALL

PROPOSED USE: RETAINING WALL TO EASTERN BOUNDARY

OVERALL SIZE : AS CALCS **No. OF STOREYS :** N/A

TYPE OF SOIL INVESTIGATION : TRIAL PITS, BOREHOLES

BEARING STRATA : WEAK MUDSTONE

TYPE OF FOUNDATION : N/A

MAXIMUM SAFE BEARING PRESSURE 200 kN/m^2

STRUCTURAL MATERIALS USED : S275 STEEL

STANDARDS AND CODES USED IN DESIGN :

General actions: Densities, self weight & imposed loads	BS EN 1991-1-1	☑
General actions: Actions on structures exposed to fire	BS EN 1991-1-2	☐
Snow Loads	BS EN 1991-1-3	☐
Wind Loads	BS EN 1991-1-4	☐
Concrete	BS EN 1992 (EC2)	☐
Steelwork	BS EN 1993 (EC3)	☑
Timber	BS EN 1995 (EC5)	☐
Masonry	BS EN 1996 (EC6)	☐
Others	CIRIA 760 GUIDENCE ON EMBEDDED RETAINING WALL DESIGN	☑
		☐
		☐

AUTHORISATION OF CALCULATION SHEETS AND BENDING SCHEDULES :

CALCULATION SHEET NOS.	CHECKED BY	DATE
Sheet nos: K01 - K14 inclusive	4P NW	21.12.23

CASE Consultants
Civil And Structural Engineers
tel: (01803) 200175
email: admin@caseconsultants.net

Calculations

Contract	NOSS ON DART DARTMOUTH	Job No.	C8829	Sheet No.	K01
		Designed by	D.P.H	Date:	Nov 2023

KING POST SUPPORT

for location refer to sheet K12 & CASE CONSULTANTS DRG No. C8829/03

These calculations cover the design of the King post and infill panels required to allow for:

Maximum retained ground level = 5.7m OD

Minimum pile cut off level = 2.5m OD

∴ Designed for a maximum excavation depth = 3.20m

Take granular made ground for 2.7m

Info taken from SOILS Ltd report 20657/FGIR Borehole B (REFER TO SHEET K14).

$\gamma b = 18kN/m^2$, $\phi = 30^\circ$ ∴ $Ka = 0.33$ & $Kp = 3.00$

Take weak silty MUDSTONE from 2.7m to 5.0m

$\gamma b = 21kN/m^2$, $\phi = 29^\circ$ ∴ $Ka = 0.35$ & $Kp = 2.88$

Take weathered MUDSTONE from 5.0m to full depth

$\gamma b = 19 kN/m^2$, $\phi = 30^\circ$, $C = 50 kN/m^3$, ∴ $Ka = 0.4$, $Kp = 2.5$, $Kpc = 3.1$

Take a surcharge of $10kN/m^2$ for 2.0m path and $15kN/m^2$ for 5.0m wide road.

Water table at 5.0m depth - Below excavation. THESE DETAILS ARE INPUT TO GROUNDFORCE SOFTWARE TO ESTABLISH WORKING PRESSURES, MOMENTS & SHEARS. (REFER TO SHEET K02 - K04 INCL.)

→ CAPPING BEAM & PERMANENT CASE TO BE CONFIRMED BY PERMANENT WORKS DESIGNER

SUMMARY

C8829/K02

INPUT

Excavation Depth	3.2 m
Surcharge	10.0 kN/m²
Active Water Depth	5.0 m
Passive Water Depth	5.0 m
Water Density	9.81 kN/m³
Min Fluid Density	5.0 kN/m³

SOIL PROFILE

Depth (m)	Soil Name	y(kN/m³)	y'(kN/m³)	C(kN/m²)	Ø(°)	Ka	Kp	Kac	Kpc	delta
0.0	MADE GROUND - Granular	18.00	11.00	0.00	30.00	0.33	3.00	0.00	0.00	0.00
2.65	Weak Slaty MUDSTONE	21.00	21.00	0.00	29.00	0.35	2.88	0.00	0.00	0.00
5.0	MUDSTONE (Weak)	19.00	20.00	50.00	30.00	0.40	2.50	1.29	3.10	0.00

SUPERIMPOSED LOADS

Type	Start m	Size kN/m²	Width m	End m	Depth m	Size kN/m²
Surcharge - Strip	2.0	5.0	5.0	0.0	0.0	0.0

SOLUTION

DESIGN SOLUTION

Support Type	Fixed Earth Support	Toe = 3.44 m
Pressure Model	BSC Piling method	
Passive Softening	Applied	
Water Balancing	Not Applied	

SHEET

Sheet Type	Z cm³/m	I cm4/m	Allowable Stress N/mm²	Allowable Moment kNm/m
LX16	1640.90	31174.80	200.00	328.20

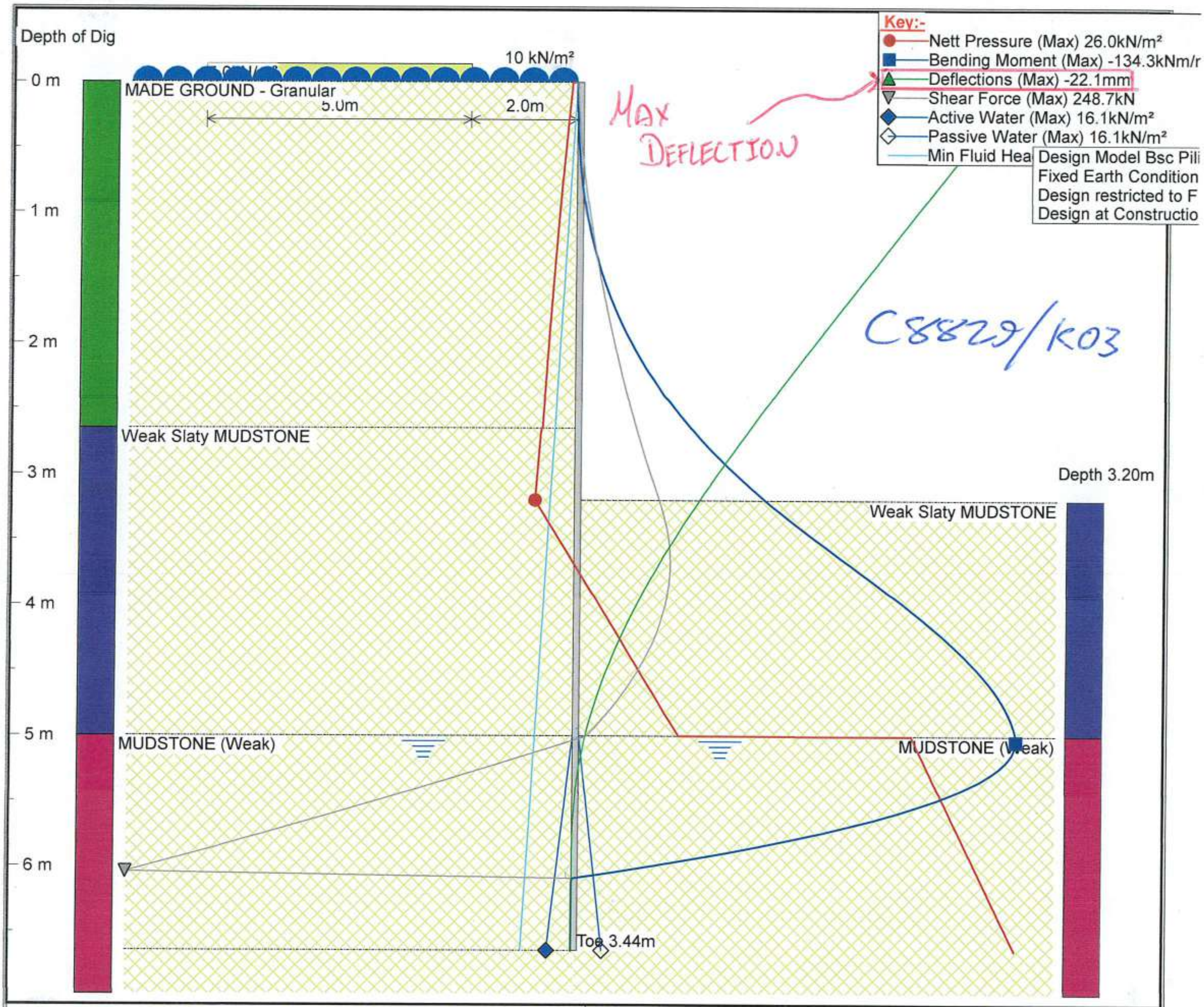
DESIGN FORCES	Maximum	Depth
Soil Pressure	26.04 kN/m²	3.2 m
Current Bending Moment	-134.26 kNm/m	5.04 m
Maximum Bending Moment	134.38 kNm/m	5.06 m
Shear Force	248.7 kN	6.04 m

Current Support Details

Depth m	Load kN/m	**GROUNDFORCE Equipment**

Maximum Support Loads

Depth (m)	Load (kN/m)	**GROUNDFORCE Equipment**



Issued for Construction.

Sheet Pile Definition

328.2kNm/m > 134.3kNm/m(Bending Capacity is Adequate)

Sheet Type:	**LX16**
Allowable Moment =	328.2 kNm/m
Moment of Inertia =	31174.8 cm4/m
Youngs Modulus (E) =	210.0 kN/mm²
Allowable Stress =	200.0 kN/mm²
Section Modulus =	1640.9 cm³/m
Pressure Model:	BSC Piling
Load Model:	N/A
Support Type:	Fixed Earth Toe-In

Groundforce
Software Licensed to:
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Title :Noss on Dart Marina

Contract :C8829
Contractor:DNA

Designer :Case Consultants
Reference:Piling mat retaining wall
Date:11/12/2023
Construction Stage Rev: 01

Groundforce Shorco
Excavation Support

C8829/K04

Depth m	Active Pressure kN/m² Pressure kN/m²	Passive Pressure kN/m² Pressure kN/m²	Bending Moment kNm/m Pressure kN/m²	Deflection mm Pressure kN/m²
0.0	3.3	0.0	0.0	-1.1
0.25	5.4	0.0	-0.1	-20.8
0.5	7.4	0.0	-0.6	-19.5
0.75	9.3	0.0	-1.5	-18.2
1.0	11.2	0.0	-3.0	-17.0
1.25	12.9	0.0	-5.2	-15.7
1.5	14.6	0.0	-8.2	-14.4
1.75	16.2	0.0	-12.2	-13.1
2.0	17.7	0.0	-17.1	-11.9
2.25	19.2	0.0	-23.1	-10.7
2.5	20.6	0.0	-30.4	-9.4
2.75	22.9	0.0	-38.9	-8.3
3.0	24.7	0.0	-48.8	-7.1
3.25	26.4	3.0	-60.3	-6.0
3.5	28.1	18.2	-73.2	-5.0
3.75	29.8	33.3	-86.7	-4.0
4.0	31.5	48.4	-100.0	-3.1
4.25	33.3	63.5	-112.2	-2.3
4.5	35.0	78.7	-122.5	-1.6
4.75	36.7	93.8	-130.1	-1.1
5.0	25.0	108.9	-134.1	-0.6
5.25	26.3	261.4	-129.8	-0.3
5.5	27.5	273.2	-110.9	-0.1
5.75	28.8	285.1	-76.7	0.0
6.0	30.0	297.0	-26.4	0.0
6.25	31.3	308.9	0.0	0.0
6.5	32.5	320.7	0.0	0.0

MAX DEFLECTION

Frame Information

MAX PRESSURE AT BOTTOM OF EXCAVATION.

MAX MOMENT

Groundforce Software Licensed to: daniel.heard	Title :Noss on Dart Marina Contract :C8829 Contractor:DNA	Designer :Case Consultants Reference:Piling mat retaining wall Date:11/12/2023 Construction Stage Rev: 01	Groundforce Shorco Excavation Support GFsafe Version 2.0.16 Copyright VP plc 2010

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Civil And Structural Engineers
tel: (01803) 200175
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Calculations

Contract	Noss on Dart Dartmouth	Job No.	C8829	Sheet No.	K05
		Designed by	D.P.H	Date:	Nov 2023

From Groundforce design software output on sheet

Max moment in trench sheet occurs at roughly 5.0m Depth with a magnitude of 1343kNm.

Max deflection = 22.1mm/m

Taking king post at 2.0m centres max moment = 134.1 × 2 = 268.2kN
max deflection = 20.8 × 2 = 41.6mm

Based on LX16 with $I_{yy} = 31175 cm^4/m$

δ Perm = $\frac{3200}{50}$ = 64mm

$\therefore I_{yy}$ Reqd = $31175 \times \frac{41.6}{64} = 20263 cm^4$

Try 254x254 UC 132 giving $I_{yy} = 22500 cm^4$.

Consider corrosion of steel at rate of 0.015mm/year/face

∴ With a design life of 60 years = 0.015x60 = 0.9mm per face.

∴ New $I_{yy} = 21100 cm^4$ considered ok as deflection is under worst case conditions

From Blue Book with $L_e = 6.64m$ & $C_1 = 1.13$

$M_R = 457kNm \gg 268 \times 1.4 = 375kNm$ (ok). ∴ considered ok with corrosion

∴ USE 254x254 UC 132 FOR KINGPOSTS @2000 CRS

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Calculations

Contract	Noss on Dart Dartmouth	Job No.	C8829	Sheet No.	K06
		Designed by	D.P.H	Date:	Nov 2023

INFILL BETWEEN PILES

Pre stressed concrete beams to be placed horizontally.

From Groundforce sheet maximum pressure at bottom of excavation = 26.4 kN/m² C (HAR)

Max Span = 2.0m.

$\therefore Md = \frac{26.4 \times 2.0^2}{8} = 13.2kNm \times 1.4 = 18.5\ kNm\ (ULT)$

Using 150mm thick 1.0m panels. ULS moment of resistance = 39.2 kNm

39.2 kNm > 18.5 kNm ∴ ok

Max shear force = $\frac{18.5 \times 2}{2}$ = 18.5 kN × 1.4 = 25.9 kN (ULT)

Using 150mm thick 1.0m panels ULS shear resistance = 82.8 kN

82.8 kN > 18.5 kN ∴ ok

∴ USE 150mm THICK × 1.0m PRE STRESSED JP CONCRETE PANELS WITH 10 NO 9.3Ø BARS

1. **Design**
   a) Concrete Beam design to BS EN 1992-1-1:2004;
   b) Permissible SLS characteristic bending moment = 24.3 kNm per 1000mm panel;
   c) ULS moment of resistance = 39.2 kNm per 1000mm panel;
   d) ULS shear resistance = 82.8 kN per 1000mm panel ;

Note: JP Concrete have designed the concrete unit only, the stability of the support conditions should be checked by overall scheme designer.

Taken from JP concrete Drawing. Prestressed Panels 150×1000 10 no 9.3 diameter.

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Calculations

Contract	NOSS ON DART DARTMOUTH	Job No.	C8829	Sheet No.	K07
		Designed by	A.C.H.	Date:	DEC23

FINAL CASE.

In the final case the pilecaps will be cast and support reinstated to +3.5m AOD.

∴ Maximum exposed height of excavation = 2.2m.

All other parameters remain as the temporary design case.

Revised maximum moment = 54.1kNm/m @ 3.7m.
(FROM SHEET K10.) = 108kNm / king post (Char).

Md ult/post = 108 × 1.4 = 151kNm.

From Previous MR for 254 × 254 UC 132 = 457kNm.

∴ residual capacity = 457 − 151
= 306kNm.

Using a standard N1 parapet system with posts @ 2.0m c/s.

Ultimate design moment = 14.1kNm with co-existent shear = 21.8kN.

Md due to parapet = 14.1 + (21.8 × 3.7) height to max moment
= 94.8kNm << 306kNm ∴ ok.

∴ 450 × 450 CONCRETE CAPING WILL BE ADEQUATE TO SUPPORT N1 PARAPET SYSTEM TO EN1317

C8829/K08

SUMMARY

INPUT

Excavation Depth	2.2 m
Surcharge	10.0 kN/m²
Active Water Depth	5.0 m
Passive Water Depth	5.0 m
Water Density	9.81 kN/m³
Min Fluid Density	5.0 kN/m³

SOIL PROFILE

Depth (m)	Soil Name	y(kN/m³)	y'(kN/m³)	C(kN/m²)	Ø(°)	Ka	Kp	Kac	Kpc	delta
0.0	MADE GROUND - Granular	18.00	11.00	0.00	30.00	0.33	3.00	0.00	0.00	0.00
2.65	Weak Slaty MUDSTONE	21.00	21.00	0.00	29.00	0.35	2.88	0.00	0.00	0.00
5.0	MUDSTONE (Weak)	19.00	12.00	50.00	30.00	0.40	2.50	1.29	3.10	0.00

SUPERIMPOSED LOADS

Type	Start m	Size kN/m²	Width m	End m	Depth m	Size kN/m²
Surcharge - Strip	2.0	5.0	5.0	0.0	0.0	0.0

SOLUTION

DESIGN SOLUTION

Support Type	Fixed Earth Support	Toe = 3.45 m
Pressure Model	BSC Piling method	
Passive Softening	Applied	
Water Balancing	Not Applied	

SHEET

Sheet Type	Z cm³/m	I cm4/m	Allowable Stress N/mm²	Allowable Moment kNm/m
LX16	1640.90	31174.80	200.00	328.20

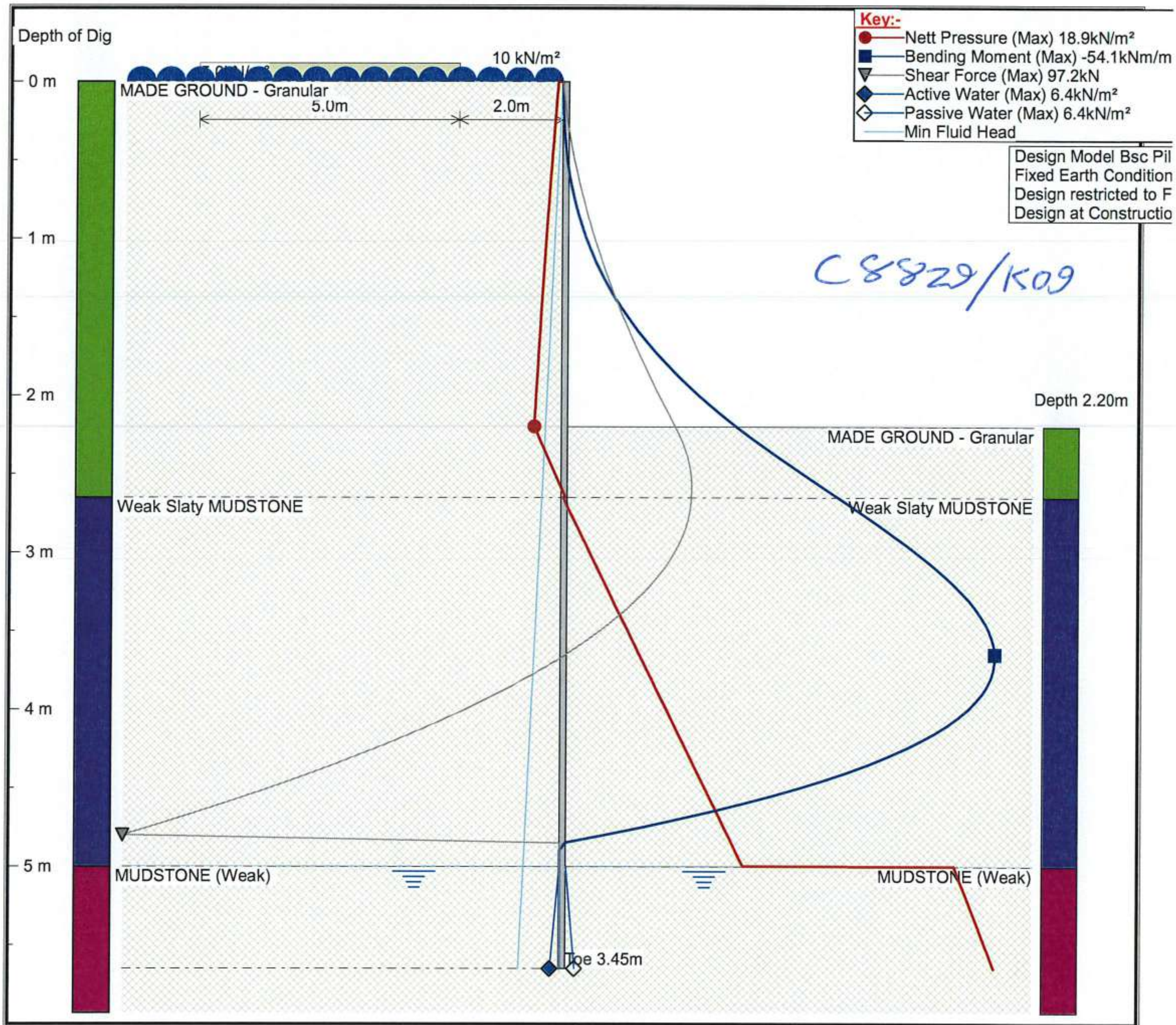
DESIGN FORCES	Maximum	Depth
Soil Pressure	18.87 kN/m²	2.2 m
Current Bending Moment	-54.12 kNm/m	3.65 m
Maximum Bending Moment	134.26 kNm/m	5.04 m
Shear Force	97.17 kN	4.8 m

Current Support Details

Depth m	Load kN/m	**GROUNDFORCE Equipment**

Maximum Support Loads

Depth (m)	Load (kN/m)	**GROUNDFORCE Equipment**



Issued for Construction.

Sheet Pile Definition

328.2kNm/m > 54.1kNm/m(Bending Capacity is Adequate)

Sheet Type:	**LX16**
Allowable Moment =	328.2 kNm/m
Moment of Inertia =	31174.8 cm4/m
Youngs Modulus (E) =	210.0 kN/mm²
Allowable Stress =	200.0 kN/mm²
Section Modulus =	1640.9 cm³/m
Pressure Model:	BSC Piling
Load Model:	N/A
Support Type:	Fixed Earth Toe-In

Groundforce
Software Licensed to:
andrew.heard@caseconsu

Title :Noss on Dart Marina

Contract :C8829
Contractor:DNA

Designer :Case Consultants
Reference:Piling mat retaining wall
Date:19/12/2023
Construction Stage Rev: 01

Groundforce Shorco
Excavation Support

C8829/K10

Depth m	Active Pressure kN/m Pressure kN/m²	Passive Pressure kN/m Pressure kN/m²	Net Pressure kN/m Pressure kN/m²	Bending Moment kNm/m Pressure kN/m²	Shear kN Pressure kN/m²	Deflection mm Pressure kN/m²
0.0	3.3	0.0	3.3	0.0	0.0	-0.3
0.1	4.2	0.0	4.2	0.0	-0.4	-5.8
0.2	5.0	0.0	5.0	-0.1	-0.8	-5.6
0.3	5.8	0.0	5.8	-0.2	-1.4	-5.4
0.4	6.6	0.0	6.6	-0.4	-2.0	-5.2
0.5	7.4	0.0	7.4	-0.6	-2.7	-5.0
0.6	8.2	0.0	8.2	-0.9	-3.5	-4.9
0.7	9.0	0.0	9.0	-1.3	-4.3	-4.7
0.8	9.7	0.0	9.7	-1.8	-5.3	-4.5
0.9	10.5	0.0	10.5	-2.3	-6.3	-4.3
1.0	11.2	0.0	11.2	-3.0	-7.3	-4.1
1.1	11.9	0.0	11.9	-3.8	-8.5	-4.0
1.2	12.6	0.0	12.6	-4.7	-9.7	-3.8
1.3	13.3	0.0	13.3	-5.8	-11.0	-3.6
1.4	13.9	0.0	13.9	-6.9	-12.4	-3.4
1.5	14.6	0.0	14.6	-8.2	-13.8	-3.3
1.6	15.2	0.0	15.2	-9.7	-15.3	-3.1
1.7	15.8	0.0	15.8	-11.3	-16.8	-2.9
1.8	16.5	0.0	16.5	-13.1	-18.5	-2.7
1.9	17.1	0.0	17.1	-15.0	-20.1	-2.6
2.0	17.7	0.0	17.7	-17.1	-21.9	-2.4
2.1	18.3	0.0	18.3	-19.4	-23.7	-2.2
2.2	18.9	0.0	18.9	-21.8	-25.5	-2.1
2.3	19.5	5.4	14.1	-24.5	-27.2	-1.9
2.4	20.0	10.8	9.2	-27.2	-28.3	-1.8
2.5	20.6	16.2	4.4	-30.1	-29.0	-1.6
2.6	21.2	21.6	-0.4	-33.0	-29.2	-1.5
2.7	22.6	26.4	-3.8	-35.9	-29.0	-1.3
2.8	23.3	32.4	-9.1	-38.8	-28.4	-1.2
2.9	24.0	38.5	-14.5	-41.6	-27.2	-1.1
3.0	24.7	44.5	-19.9	-44.2	-25.5	-0.9
3.1	25.4	50.6	-25.2	-46.7	-23.2	-0.8
3.2	26.0	56.6	-30.6	-48.8	-20.4	-0.7
3.3	26.7	62.7	-35.9	-50.7	-17.1	-0.6
3.4	27.4	68.7	-41.3	-52.2	-13.2	-0.5
3.5	28.1	74.8	-46.7	-53.3	-8.8	-0.4
3.6	28.8	80.8	-52.0	-54.0	-3.9	-0.4
3.7	29.5	86.9	-57.4	$\boxed{-54.1}$	1.6	-0.3
3.8	30.2	92.9	-62.8	-53.6	7.6	-0.2
3.9	30.9	99.0	-68.1	-52.5	14.1	-0.2
4.0	31.5	105.0	-73.5	-50.8	21.2	-0.1
4.1	32.2	111.1	-78.9	-48.3	28.8	-0.1
4.2	32.9	117.2	-84.2	-45.0	37.0	-0.1
4.3	33.6	123.2	-89.6	-40.9	45.7	0.0
4.4	34.3	129.3	-95.0	-35.8	54.9	0.0
4.5	35.0	135.3	-100.3	-29.8	64.7	0.0
4.6	35.7	141.4	-105.7	-22.9	75.0	0.0
4.7	36.4	147.4	-111.0	-14.8	85.8	0.0
4.8	37.1	153.5	-116.4	-5.7	97.2	0.0
4.9	37.8	159.5	-121.8	0.0	0.0	0.0
5.0	25.0	165.6	-140.6	0.0	0.0	0.0
5.1	25.5	303.4	-277.9	0.0	0.0	0.0
5.2	26.0	308.1	-282.1	0.0	0.0	0.0
5.3	26.5	312.9	-286.4	0.0	0.0	0.0
5.4	27.0	317.6	-290.6	0.0	0.0	0.0
5.5	27.5	322.4	-294.9	0.0	0.0	0.0
5.6	28.0	327.1	-299.1	0.0	0.0	0.0

Frame Information

Groundforce
Software Licensed to:
andrew.heard@caseconsu

Title :Noss on Dart Marina

Contract :C8829
Contractor:DNA

Designer :Case Consultants
Reference:Piling mat retaining wall
Date:19/12/2023
Construction Stage Rev: 01

Groundforce Shorco
Excavation Support

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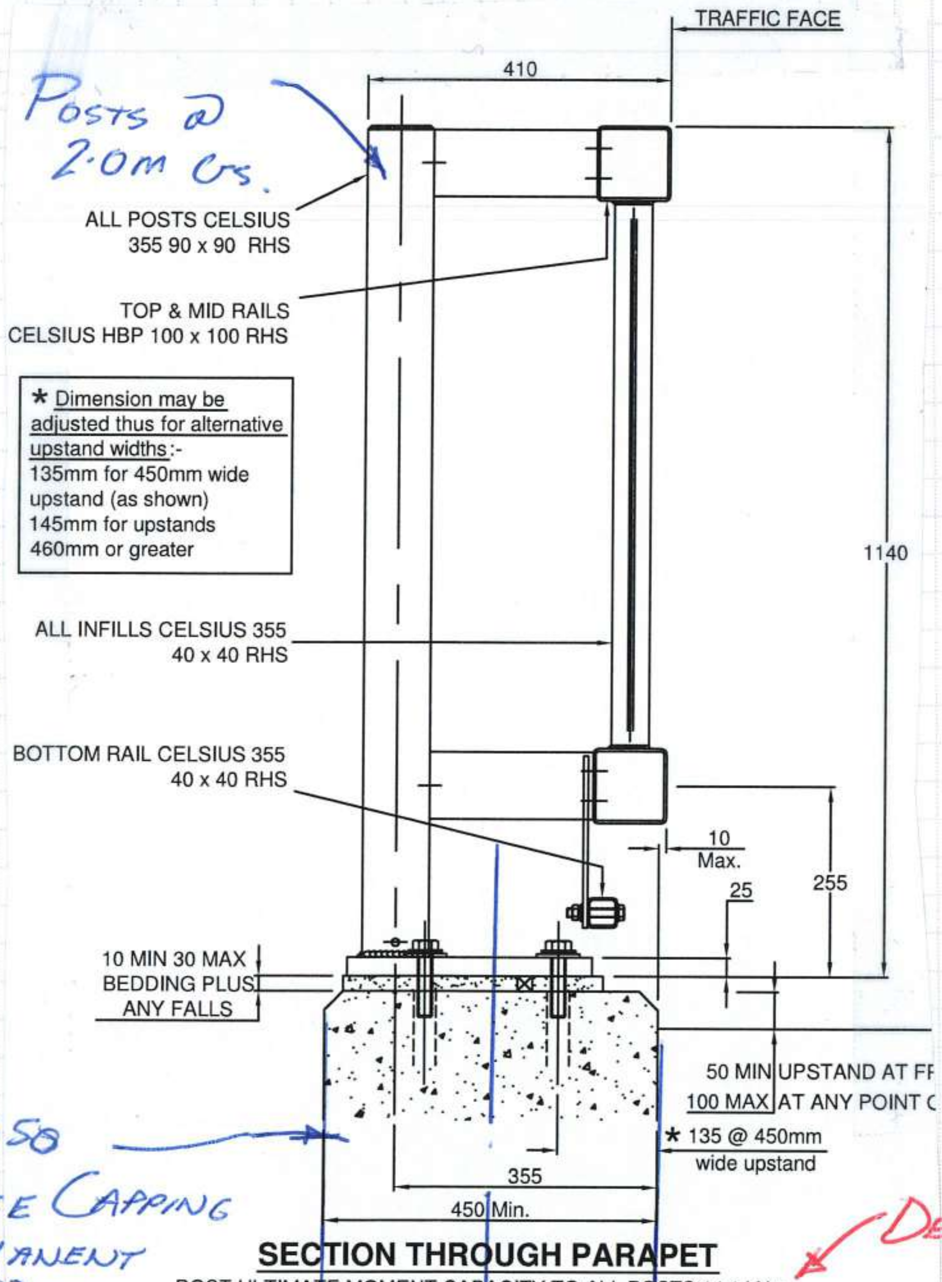
tel: (01803) 200175
email: admin@caseconsultants.net

Calculations

Contract	Noss on Dart. Dartmouth	Job No.	C8829	Sheet No.	K11
		Designed by	a.C.H.	Date:	Dec23

Posts @ 2.0m C/s.



450x450
Concrete Capping
By Permanent
Works Designer.

Design Loads.

SECTION THROUGH PARAPET

POST ULTIMATE MOMENT CAPACITY TO ALL POSTS 14.1 kNm,
COEXISTING SHEAR 21.8 kN,
POST ULTIMATE SHEAR CAPACITY 131 kN

℄ of Pile.

Parapet Loadings.

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Calculations

Contract	Noss on Dart Dartmouth	Job No.	C8829	Sheet No. K12
		Designed by	D.P.H	Date: Nov 2023.

PLAN ON KINGPOST WALL

1:250.

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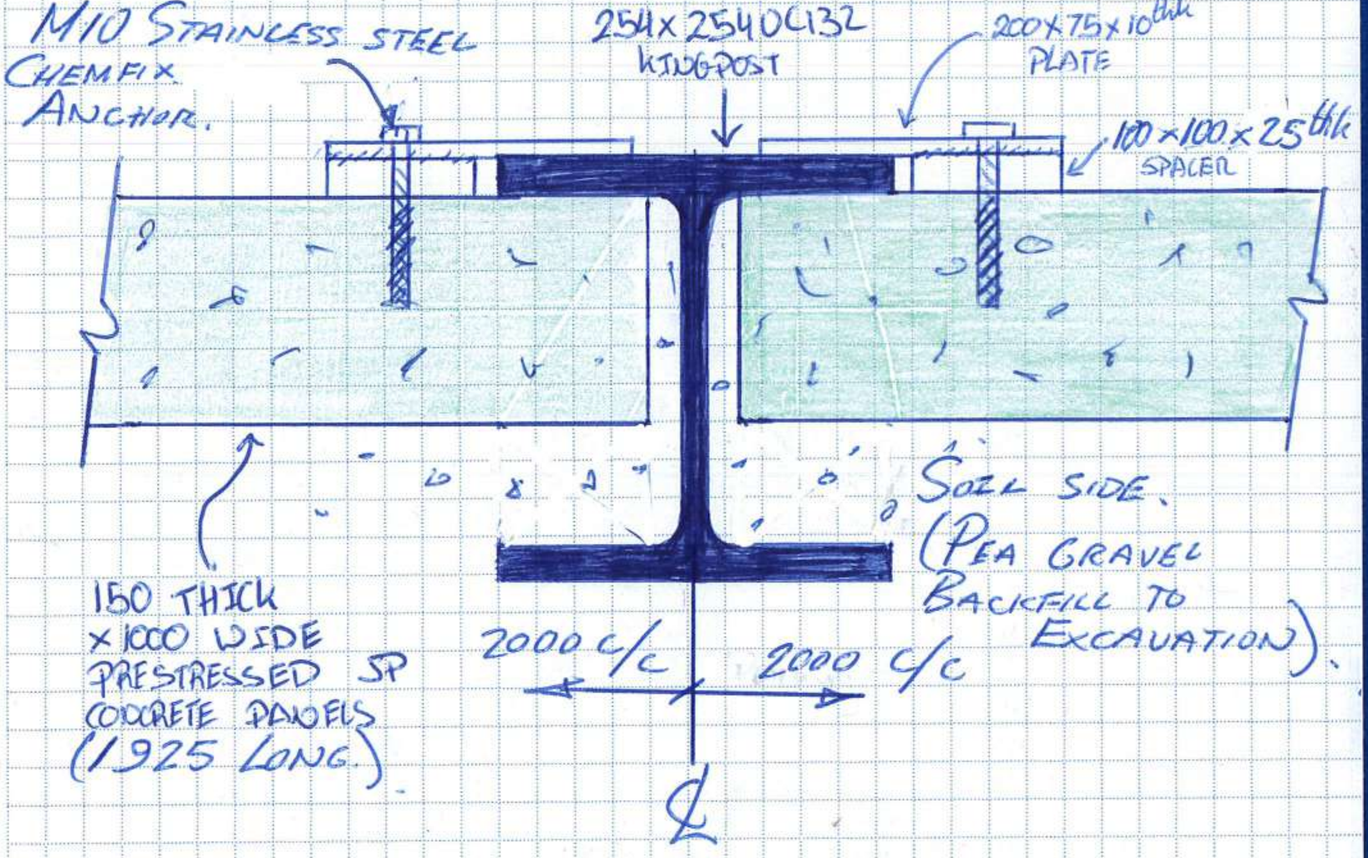
Calculations

Contract	NOSS ON DART DARTMOUTH	Job No.	C8829	Sheet No.	K13
		Designed by	D.P.H	Date:	Nov 2023



PLAN OF KINGPOST

4No LOCATION PLATES PER PANEL.

soils LIMITED

Contract Name:	Noss-on-Dart Marina	Client:	Premier Marinas Ltd	Hole ID:	BHB
Contract Number: 20652	Date Started: 12/01/2023	Logged By: DW	Checked By: SB	Status: FINAL	Hole Type: RC
Easting:	Northing:	Ground Level:	Plant Used: Camacchio 205	Print Date: 24/02/2023	Scale: 1:50

Weather:

Termination: Sufficient depth into bedrock achieved.

SPT Hammer: ADS R2 Energy Ratio: 63%

Sheet 1 of 2

C 8829/K14

Samples & In Situ Testing

Depth	Tests/Samples	TCR	SCR	RQD	FI	Water Added	Level (mAOD)
1.20	S						
1.70	D						
1.60 - 1.90		0	0	0			
2.40	D						
1.90 - 3.00		100	0	0			
2.70	D						
3.20	PL						
3.00 - 3.90		44	0	0			
3.70	D						
3.90	100 (25 for 75mm/100 for 177mm)(S) S						
3.90 - 4.65		67	0	0			
4.70	D						
4.65 - 5.30		92	0	0			
5.00	D						
5.30 - 6.00		100	0	0			
5.70	D						
6.20	D						
6.00 - 6.90		100	0	0			
6.70	D						
7.20	D						
6.90 - 7.70		75	0	0			
7.60	D						
7.95	PL						
7.70 - 8.60		78	0	0			
8.70	D						
8.60 - 8.90		33	0	0			
8.90 - 9.50		100	0	0			
9.20	D						
9.70	D						

CONSERVATIVELY TAKE AS MADE GROUND TO THIS LEVEL.

WEAK MUDSTONE

MUDSTONE

Strata Details

Depth (m) (Thickness)	Legend	Strata Description
(1.20) 1.20		Driller's description. MADE GROUND.
(0.45) 1.65		Light brown mottled multicoloured, clayey GRAVEL. Gravel is angular fine to medium weathered extremely weak slaty mudstone and quartzite. WEATHERED BOVISAND FORMATION
(0.70) 2.35		Light brown mottled multicoloured, silty/clayey GRAVEL/gravelly SILT. Gravel is angular fine to medium weathered extremely weak slaty mudstone. WEATHERED BOVISAND FORMATION
(0.30) 2.65		Blueish grey mottled dark grey and light brown, slightly silty/clayey GRAVEL. Gravel is angular fine to medium weathered extremely weak slaty mudstone. WEATHERED BOVISAND FORMATION
(2.00) 4.65		Dark grey thinly laminated grey, extremely weak silty slaty MUDSTONE. Recovered as angular fine to coarse gravel and cobble sized fragments with rare quartzite veins. BOVISAND FORMATION
(3.05) 7.70		Dark grey and grey thinly laminated and cross bedded, slaty mudstone. Intact fragments are medium strong. Occasional quartzite seams in line with and perpendicular to bedding planes. Frequent zones of folding. Predominantly recovered as angular fine to coarse gravel and cobbles. BOVISAND FORMATION Quartzite cobble.
(1.20) 8.90		Dark grey and grey thinly laminated and cross bedded, slaty mudstone. Intact fragments are medium strong. Occasional quartzite seams in line with and perpendicular to bedding planes. Predominantly recovered as angular fine to coarse gravel and cobbles. BOVISAND FORMATION Fragmented quartzite cobble.
		Dark grey with rare thin grey laminations, silty slaty MUDSTONE. Recovered as thinly laminated full diameter extremely weak non-bonded fragments. BOVISAND FORMATION Fragmented quartzite cobble.

PERCHED WATER.
-ALL OTHER AREAS >>5.0m
(NOW DISAPATED.)

Groundwater: Water Strike | Backfill/Installation

Start & End of Shift Observations

Date	Time	Depth (m)	Casing (m)	Water (m)

Borehole Diameter: Depth (m)	Borehole Diameter: Dia (mm)	Casing Diameter: Depth (m)	Casing Diameter: Dia (mm)

Remarks:
Water flush used.
No roots observed

Flush Information

Top (m)	Base (m)	Flush Type	Return	Flush Colour

Installation

Top (m)	Base (m)	Type	Dia (mm)

Water Strikes

Strike (m)	Casing (m)	Sealed (m)	Time (mins)	Rose to (m)	Remarks
1.20	1.20		20	0.82	

Fracture spacing reported in mm as Min, Average and Max values. TCR, SCR and RQD reported in %. Hand vane (HV) reported in kPa.